

Material Test Reports – Summary

Independent verification by SGS — chemical & mechanical properties

REPORT NO. SGS-KM-2024-0125	SAMPLED 2024-01-18	LAB SGS Tianjin
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1. Chemical Composition (%) – Wire Raw Material

Grade	C	Si	Mn	P ≤	S ≤	Cr	Ni
Q195 (spec)	≤0.12	≤0.30	≤0.50	0.035	0.040	—	—
Q195 (tested)	0.09	0.21	0.42	0.021	0.028	—	—
SS304 (spec)	≤0.08	≤1.00	≤2.00	0.045	0.030	18–20	8–10.5
SS304 (tested)	0.05	0.46	1.28	0.028	0.015	18.3	8.2
SS316 (spec)	≤0.08	≤1.00	≤2.00	0.045	0.030	16–18	10–14
SS316 (tested)	0.04	0.51	1.35	0.024	0.012	16.8	10.6

2. Mechanical Properties – Finished Wire

Test	Method	Requirement	Result	Verdict
Tensile strength (Q195)	ASTM A370	400–550 MPa	468 MPa	Pass
Elongation (Q195)	ASTM A370	≥ 12%	18%	Pass
Tensile strength (SS304)	ASTM A370	≥ 515 MPa	601 MPa	Pass
Zinc coating mass	ISO 1460	≥ 80 μm equiv.	86 μm	Pass
Weld shear (4 mm)	ASTM A1064	≥ 50% wire load	63%	Pass
Salt spray 1,000 h	ASTM B117	No red rust	No red rust	Pass

3. Sampling & Statement

Samples drawn at random from production lots 2401-A through 2401-F by SGS inspector at plant. Full traceability maintained from coil to finished batch. Test certificates for specific order lots are supplied with each shipment.

This is a summary for reference. Complete SGS reports including instrument calibration data are available under NDA for qualified projects.